Anmol Patalay

A1-801 Gagan Lawish, Pisoli, Pune

+91 9834545186 | anmolpatalay@gmail.com | LinkedIn | GitHub

Education

Maharashtra Institute of Technology, WPU - CGPA: 9.26/10 **Jul 2021 – Jul 2025**

Bachelor of Technology in Electronics and Communication Engineering (AI & ML Specialization) *Pune, Maharashtra*

Professional Summary

Software Engineer with experience building Python backend systems, automation frameworks, and AI-powered applications. Skilled in FastAPI, SQL, Docker, REST APIs, and scalable backend development. Reduced manual testing effort by 70% through automation and passionate about building production-grade software systems.

Experience

Visteon Corporation **August 2025 – current**

Software Engineer *Pune, Maharashtra*

- Build a Python based backend automation framework that reduces manual testing effort and time by 70%.
- Collaborated with software and system teams to analyze complex issues, perform root cause analysis and deliver technical solutions.
- Developed reusable Python modules for hardware-software integration and automated validation workflows.
- Built a pipeline using FastAPI for getting live Execution data using Python listeners for getting real time data.

Projects

Secure Job Application Tracking Backend GitHub **Jul 2026 – Aug 2026**

- Built a RESTful backend with 20+ endpoints using FastAPI, SQLAlchemy ORM, MySQL, and Pydantic for schema validation and API contract enforcement.
- Implemented JWT authentication (OAuth2 Password Flow), bcrypt password hashing, dependency-injected authorization guards, and user-scoped access control.
- Developed secure account management workflows including registration, login, password updates, and protected profile access.
- Implemented request rate limiting to mitigate brute-force attacks and improve API security.
- Designed the project using modular architecture and reusable service layers to support future scalability and maintainability.

Production-Ready House Price Prediction API Live Demo **Aug 2026**

- Built and trained a machine learning regression model for California house price prediction and serialized it using Joblib for efficient inference.
- Developed a FastAPI backend exposing prediction functionality through REST endpoints for real-time inference.
- Containerized the application using Docker and optimized image builds using dependency-layer caching for faster deployments.
- Deployed the service on Render through GitHub-integrated deployment workflows, enabling public API accessibility.
- Implemented input validation and structured API responses to improve reliability and user experience.

AI in Agriculture: Grape Cluster Segmentation **2023**

- Developed a U-Net-based semantic segmentation model for grape-cluster detection using the GrapesNet dataset in collaboration with IIT Hyderabad.
- Trained and optimized the model on NVIDIA T4 GPUs using TensorFlow/Keras, data augmentation, Binary Cross-Entropy loss, and hyperparameter tuning.
- Achieved 97.93% Pixel Accuracy, 94.87% Dice Score, and 90.25% IoU, demonstrating strong segmentation performance.
- Leveraged TensorFlow, OpenCV, NumPy, CUDA, and computer vision techniques to build a high-performance research solution.

Technical Skills

Languages: Python, C++, SQL
AI/ML: scikit-learn, TensorFlow, PyTorch, U-Net, CNN, Lang Chain, HuggingFace Transformers, OpenAI API, RAG pipelines, Prompt Engineering
Backend/APIs: FastAPI, Flask, REST APIs, SQLAlchemy, MySQL, Redis (basics)
SQL/Databases: MySQL, Relational Database Design and Internals, Complex Joins, CTEs, Window Functions, Views, Normalization, Query Optimization, Indexing, Transactions (ACID), System design
Devops/Tools: Docker, Git, GitHub Actions (CI/CD), Linux CLI